

Vijayasree Paluvoi

Data Engineer

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PROFESSIONAL SUMMARY

Data Engineer with **5+ years** of experience designing and optimizing **scalable data pipelines**, **real-time streaming architectures**, and **cloud-native data platforms** across **GCP, Azure, and AWS**. Proven expertise in **Apache Spark, PySpark, Kafka, Databricks, Delta Lake**, and **Snowflake**, processing **100M+ events/hour** and large-scale batch workloads. Strong background in **ETL/ELT development**, **data modeling**, and **distributed systems**, with hands-on experience building **data lakes**, **Medallion architectures**, and **ML/AI data pipelines**. Adept at improving **performance**, **data quality**, and **pipeline reliability**, enabling **data-driven decision-making** and **advanced analytics**.

TECHNICAL SKILLS

Programming & Query Languages: Python (PySpark, Pandas, NumPy), SQL, Bash (Shell Scripting), PowerShell

Big Data & Streaming: Apache Spark, PySpark, Apache Kafka, Structured Streaming, Databricks, Delta Lake

Data Engineering & ETL: ETL/ELT Pipelines, Batch Processing, Stream Processing, Apache Airflow, Azure Data Factory, dbt, Schema Registry, Medallion Architecture (Bronze/Silver/Gold), Data Partitioning, Data Compaction

Cloud Platforms: AWS (S3, EC2, IAM, VPC, CloudWatch), Azure (Data Factory, Databricks, Synapse Analytics, ADLS Gen2, Event Hubs), GCP (BigQuery, Cloud Storage, Cloud Composer, GKE)

Data Warehousing & Databases: Snowflake, BigQuery, Azure Synapse Analytics, PostgreSQL, SQL Server, MySQL

Data Storage & Formats: Amazon S3, Azure Data Lake Storage Gen2, Google Cloud Storage, Delta Lake, Parquet, Avro, JSON

Machine Learning & AI Integration: MLflow, Azure OpenAI Service, LangChain, Prompt Flow, Azure OpenAI Service, LangChain, Hugging Face Transformers, PyTorch, Scikit-learn, RAG (Retrieval-Augmented Generation), Vector Databases (FAISS, Pinecone), NLP, MLflow, Feature Engineering

Data Quality & Governance: Great Expectations, Azure Purview, Data Lineage, Metadata Management, Schema Evolution, Data Validation, Data Observability, SLA Monitoring

DevOps & Infrastructure: Docker, Kubernetes, Terraform, Helm, CI/CD, Azure DevOps, Git, GitHub Actions, IaC, ARM Templates

Visualization & Analytics: Power BI, Tableau

Methodologies & Architecture: Agile, Scrum, DataOps, SDLC, Distributed Systems, Event-Driven Architecture

PROFESSIONAL EXPERIENCE

NVIDIA | CA | Data Engineer II

September 2023 – Present

- Designed and maintained **real-time data pipelines** processing **100M+ events/hour** from **10K+ nodes** using **Apache Kafka, PySpark Structured Streaming, Kubernetes**, enabling **low-latency monitoring** of DGX Cloud infrastructure.
- Developed **stream processing applications** on **GCP** using **Kafka** and **PySpark**, implementing **windowing, watermarking, and late data handling** across **40+ telemetry streams**, improving **data reliability** and **fault tolerance**.
- Integrated **NVIDIA DCGM telemetry metrics** into **Kafka ingestion pipelines**, capturing **GPU utilization, memory, and thermal data** to enhance **observability for ML/AI workloads**.
- Optimized **Apache Spark workloads** by tuning **shuffle partitions, executor memory, and join strategies**, supporting **4K+ metrics/sec ingestion per cluster** and improving **processing performance and scalability**.
- Contributed to building a **petabyte-scale data lake** using **Delta Lake, Google Cloud Storage (GCS)**, and **BigQuery**, enabling **LLM training datasets, data versioning, and deduplication pipelines**.
- Built and orchestrated **Apache Airflow (Cloud Composer) DAGs** for **data ingestion, partitioning, and Delta Lake optimization**, improving **pipeline reliability** and **scheduling efficiency**.
- Developed **Snowflake data pipelines** using **Snowpipe** and **dbt**, integrating **MLflow** and **Weights & Biases** experiment data for **analytics and reporting**.
- Created **analytics dashboards** using **SQL, PySpark, and Snowflake** to analyze **GPU utilization**, driving **cost optimization and resource efficiency**.

OpenText | ON | Data Engineer

December 2022 – September 2023

- Designed and maintained **scalable batch data pipelines** on **Microsoft Azure** using **Azure Data Factory**, **Azure Synapse Analytics**, **Apache Spark**, and **Medallion Architecture** on **ADLS Gen2**, processing **80GB+ daily** and reducing **ETL runtime by 20%**.
- Built **event-driven and near real-time data pipelines** using **Azure Event Hubs**, **Azure Functions**, **Azure Service Bus**, and **API Management**, enabling **cross-cloud data integration (Azure + AWS)**.
- Developed and optimized **distributed data processing pipelines** using **Python**, **PySpark**, and **Databricks**, reducing **batch processing time from 2 hours to 40 minutes** and improving **pipeline efficiency**.
- Implemented **data enrichment and transformation workflows** using **Azure OpenAI Service**, **LangChain**, and **PySpark**, enabling **large-scale unstructured data processing** and improving **data classification accuracy** and **analytics usability**.
- Established **data quality and validation frameworks** using **Great Expectations**, **SQL**, **Azure Monitor**, **Application Insights**, and **Log Analytics**, reducing **data issue detection and resolution time by 30%**.
- Automated **CI/CD pipelines** and **infrastructure deployments** using **Azure DevOps**, **Git**, **Terraform**, and **ARM templates**, improving **deployment reliability** and **release velocity**.
- Mentored **junior engineers** on **PySpark optimization**, **dbt data modeling**, and **data engineering best practices**, increasing **team productivity** and **code quality**.

Microsoft | ON | Data Engineer

July 2020 – December 2022

- Designed and implemented **end-to-end data pipelines** using **Azure Data Factory (ADF)**, **Azure Synapse Analytics**, **Azure Databricks**, **Delta Lake**, and **Medallion Architecture (Bronze/Silver/Gold)** on **ADLS Gen2** and **Azure Blob Storage**.
- Optimized **ETL/ELT pipelines** and **cloud resource utilization**, reducing **processing time by 20%** while enhancing **data governance**, **metadata management**, and **lineage tracking** using **Azure Purview**.
- Built **batch** and **near real-time data ingestion pipelines** using **Azure Event Hubs**, **Azure Stream Analytics**, **Databricks Structured Streaming**, and **Azure Synapse**, enabling **scalable analytics** and **ML workloads**.
- Developed **LLM-based data enrichment solutions** using **Azure OpenAI Service** and **Prompt Flow**, improving **data accuracy** and **analytical insights by 15%**.
- Engineered **feature engineering** and **ML data pipelines** using **Python**, **PySpark**, **SQL**, and **scikit-learn**, integrated with **Azure Machine Learning (Azure ML)** for **model training**, **deployment**, and **inference**.
- Implemented **data quality**, **monitoring**, and **CI/CD frameworks** using **SQL**, **Azure Monitor**, **Application Insights**, **ARM templates**, and **Azure DevOps**, reducing **incident resolution time by 25%** and improving **system reliability**.

Manulife | ON | Cloud Engineer

August 2018 – May 2020

- Designed and maintained **cloud infrastructure** on **Microsoft Azure** and **AWS** using **Terraform** and **ARM templates (Infrastructure as Code - IaC)**, reducing **environment provisioning time by 30%** and enabling **scalable, repeatable deployments**.
- Supported **migration of on-premises systems to Azure**, including **SQL Server to Azure SQL Database** and **file storage to Azure Blob Storage**, ensuring **successful testing**, **validation**, and **production cutovers**.
- Implemented **Identity and Access Management (IAM)** using **Azure Active Directory**, **Azure Key Vault**, and **AWS IAM**, enforcing **RBAC policies**, **secure secrets management**, and **compliance standards**.
- Configured **hybrid cloud networking** using **Azure VNet**, **VPN Gateway**, and **AWS VPC**, enabling **secure connectivity between on-prem and cloud environments**.
- Built **monitoring, logging, and alerting solutions** using **Azure Monitor** and **AWS CloudWatch**, and automated operational workflows using **PowerShell**, reducing **manual effort by 25%**.
- Collaborated with **DevOps** and **engineering teams** to support **CI/CD pipelines (Azure DevOps)**, participated in **on-call rotations**, and developed **runbooks/SOPs** to improve **incident response** and **operational efficiency**.

EDUCATION

MS in Information, Network & Computer Security | New York Institute of Technology, Vancouver, CANADA